

SUMMARY SHEET

Summary Sheet

Conference metadata, abstract, AI-use disclosure, and package contents

Author: Ravi Bajaj | **Target venue:** ML4H 2026 Proceedings Track | **Course:** AI in Healthcare

Conference / Symposium

Field	Value
Name	Machine Learning for Health Symposium (ML4H 2026), Proceedings Track
Conference URL	https://ml4h.ahli.cc/
Author instructions / CFP	https://ml4h.ahli.cc/submit/call-for-papers/
Submission deadline	September 10, 2026, 11:59 PM AoE
Author notification	October 22, 2026
Camera-ready deadline	November 7, 2026 (tentative)
Event dates	December 6-7, 2026
Location	Sydney, Australia
Submission type	Proceedings paper: up to 8 pages at submission, excluding references and appendices; accepted papers are published in PMLR.

Final Paper Title and Abstract

Title: Surrogate-Aware Runtime Assurance for Agentic Medical LLMs

Abstract: Medical large language models (LLMs) are often evaluated for hallucination and citation accuracy, but agentic clinical workflows introduce a subtler failure: evidence overclaiming by systems that retrieve, summarize, plan, and draft clinical-sounding actions. This paper proposes a runtime-assurance framework that gates LLM outputs and proposed actions by citation validity, endpoint strength, study design, population fit, actionability, opportunity cost, clinician authority, source provenance, runtime budget order, confounding, proportional exposure, witness selection, privacy loss, and consent aggregation. A synthetic stress-test set distinguishes hard outcomes from surrogate/process endpoints, flags unverifiable citations, treats low clinician uptake as possible safety filtering, and requires abstention when context is missing. Added probes cover recurring hospital genetics triage, including pharmacogenomic alerts, newborn-screening follow-up, ACMG secondary findings, hereditary cancer/FH/Lynch flags, VUS handling, and family-linked privacy boundaries, plus multimodal telemedicine, wearable physiology, cross-script provenance,

sequential depletion, orthogonal projection, Kelly-style exposure limits, uniform witness sampling, and family genomic consent under differential-privacy-style budget accounting. It does not validate a clinical product, but provides a testable architecture for safer clinician-facing agentic medical LLMs.

Generative AI Disclosure

Generative AI was used to assist with brainstorming, literature organization, code scaffolding, document drafting, revision, and formatting. The author reviewed, directed, and revised the conceptual framing; selected the capstone direction; provided the course materials, prior assignment insights, and instructor feedback; and remains responsible for all claims, citations, code, and final submissions. No real patient data were used. The synthetic stress tests do not provide medical advice, diagnosis, or treatment recommendations.

Optional Data, Code, and Tools

- Deterministic evidence-gate stress-test harness with 30 synthetic scenarios.
- ZDD-style sparse-family audit over 39 evidence features.
- Model-selection audit over 19 candidate claim forms.
- Stage 1 probes: depletion ordering, orthogonal projection, Kelly exposure, uniform witnesses, consent/privacy budget, finite event algebra, measure-on-measures, and Mahalanobis covariate distance.
- Consent audit blocks full raw genomic release while preserving partial inclusion; Mahalanobis flags `genomeindia_like`, `latin_america_admixed_like` for source-specific validation.
- CSV/JSON results, reproducible Python tests, generated figures, and one demo video are included; all are synthetic or analogy-only, not clinical evidence.